

New Wheels Project

Introduction to SQL

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

By Javeria Siddiq

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

```
SELECT COUNT(DISTINCT customer_id) AS 'Total_Customers'
FROM order_t;
```

```
SELECT state, COUNT(customer_id) AS 'Total_Customers',
COUNT(customer_id) * 100 / (SELECT COUNT(*) FROM customer_t) AS 'Percentage'
FROM customer_t
GROUP BY state
ORDER BY COUNT(customer_id) DESC;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT COUNT(DISTINCT customer_id) AS 'Total_Customers'
FROM order_t
```

Output:

Showing 1 rows

Total_Customers
994

Result: Passed

Query 1

Query:

```
SELECT state, COUNT(customer_id) AS 'Total_Customers',
COUNT(customer_id) * 100 / (SELECT COUNT(*) FROM customer_t) AS 'percentage'
FROM customer_t
GROUP BY state
ORDER BY COUNT(customer_id) DESC
```

Output:

Showing first 10 rows out of 49 rows

state	Total_Customers	percentage
Texas	97	9
California	97	9
Florida	86	8
New York	69	6
District of Columbia	35	3

Output:

Showing first 10 rows out of 49 rows

state	Total_Customers	percentage
Texas	97	9
California	97	9
Florida	86	8
New York	69	6
District of Columbia	35	3
Ohio	33	3
Colorado	33	3
Alabama	29	2
Washington	28	2
Arizona	26	2

Observations and Insights:

- There are 994 total customers who have placed orders.
- From the chart above, it looks like Texas, California, Florida and New York have the highest number of customers, totaling to be 349 customers out of 994 total customers, which is about 32% of all the customers who have purchased cars from these states in total.
- States like District of Columbia, Ohio, and Colorado make around 101 customers, which is about 9% of total customers in these states, which is in the mid-range.
- As seen by the chat above, excluding the 4 states that have the largest customers and the mid-range states, 59% of the customers come from states that have less total customers, but makes the large portion of the customers too.

Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
SELECT p.vehicle_maker,
COUNT(DISTINCT o.customer_id) AS 'Total_Customers'
FROM product_t AS 'p'
INNER JOIN order_t AS 'o'
ON p.product_id = o.product_id
GROUP BY p.vehicle_maker
ORDER BY COUNT(DISTINCT o.customer_id) DESC
LIMIT 5;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT p.vehicle_maker,
COUNT(DISTINCT o.customer_id) AS 'total_customers'
FROM product_t AS 'p'
INNER JOIN order_t AS 'o'
ON p.product_id = o.product_id
GROUP BY p.vehicle_maker
ORDER BY COUNT(DISTINCT o.customer_id) DESC
LIMIT 5
```

Output:

Showing 5 rows

vehicle_maker	total_customers
Chevrolet	83
Ford	63

Output:

Showing 5 rows

vehicle_maker	total_customers
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50

Observations and Insights:

- Top 5 vehicle makers preferred by the customers are Chevrolet, Ford, Toyota, Pontiac, and Dodge.
- Around 298 total customers purchase these 5 vehicles, which is around 30% of the total customer base (298 / 994)
- However, 70% of the total customer base comes from the other vehicle makers, that needs to be considered as well.

Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
SELECT *
FROM
(SELECT p.vehicle_maker, c.state,
      COUNT(DISTINCT c.customer_id) AS 'Total_customers',
      RANK() OVER(PARTITION BY c.state ORDER BY COUNT(DISTINCT c.customer_id)
      DESC) AS 'rnk'
FROM product_t AS 'p'
INNER JOIN order_t AS 'o'
on p.product_id = o.product_id
INNER JOIN customer_t AS 'c'
on c.customer_id = o.customer_id
GROUP BY p.vehicle_maker, c.state) AS 'New_Table'
WHERE rnk = 1;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT *
FROM
(SELECT p.vehicle_maker, c.state,
      COUNT(DISTINCT c.customer_id) AS 'Total_customers',
      DENSE_RANK() OVER(PARTITION BY c.state ORDER BY COUNT(DISTINCT c.customer_id) DESC) AS 'rnk'
FROM product_t AS 'p'
INNER JOIN order_t AS 'o'
on p.product_id = o.product_id
INNER JOIN customer_t AS 'c'
on c.customer_id = o.customer_id
GROUP BY p.vehicle_maker, c.state) AS 'New_Table'
WHERE rnk = 1
```

Output:

Showing first 10 rows out of 143 rows

vehicle_maker	state	Total_customers	rnk
Dodge	Alabama	5	1

Showing first 10 rows out of 143 rows

vehicle_maker	state	Total_customers	rnk
Dodge	Alabama	5	1
Chevrolet	Alaska	2	1
Cadillac	Arizona	3	1
Pontiac	Arizona	3	1
Chevrolet	Arkansas	1	1
GMC	Arkansas	1	1
Mitsubishi	Arkansas	1	1
Pontiac	Arkansas	1	1
Suzuki	Arkansas	1	1
Volkswagen	Arkansas	1	1

Observations and Insights:

- From the data presented above, it looks like Dodge is the top choice for the customers in Alabama.
- In Arizona, both Cadillac and Pontiac are the top choices for the customers.
- As seen above, in Arkansas there are 6 different vehicles that are preferred by the customers, which are Chevrolet, GMC, Mitsubishi, Pontiac, Suzuki, and Volkswagen as they all have the 1st rank.

Question 4: Find the overall average rating given by the customers. What is the average rating in each quarter?

Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

Solution Query:

```
SELECT AVG(Feedback_Rating) AS 'Avg_Rating'
FROM
    (SELECT customer_id, customer_feedback,
        CASE customer_feedback
            WHEN 'Very Bad' THEN 1
            WHEN 'Bad' THEN 2
            WHEN 'Okay' THEN 3
            WHEN 'Good' THEN 4
            WHEN 'Very Good' THEN 5
        END AS 'Feedback_Rating'
    FROM order_t)
AS 'Rating_table';

SELECT quarter_number,
    Round(AVG(Feedback_Rating), 2) AS 'Avg_Rating_Per_Quarter'
FROM
    (SELECT quarter_number,
        CASE customer_feedback
            WHEN 'Very Bad' THEN 1
            WHEN 'Bad' THEN 2
            WHEN 'Okay' THEN 3
            WHEN 'Good' THEN 4
            WHEN 'Very Good' THEN 5
        END AS 'Feedback_Rating'
    FROM order_t) AS 'Rating_table'
GROUP BY quarter_number;
```




Output:

Result: Passed

Query 1

Query:

```
SELECT AVG(Feedback_Rating) AS 'Avg_Rating'
FROM
  (SELECT customer_id, customer_feedback,
    CASE customer_feedback
      WHEN 'Very Bad' THEN 1
      WHEN 'Bad' THEN 2
      WHEN 'Okay' THEN 3
      WHEN 'Good' THEN 4
      WHEN 'Very Good' THEN 5
    END AS 'Feedback_Rating'
  FROM order_t)
AS 'Rating_table'
```

Output:

Showing 1 rows

Avg_Rating
3.135

Result: Passed

Query 1

Query:

```
SELECT quarter_number,
Round(AVG(Feedback_Rating), 2) AS 'Avg_Rating_Per_Quarter'
FROM
  (SELECT quarter_number,
    CASE customer_feedback
      WHEN 'Very Bad' THEN 1
      WHEN 'Bad' THEN 2
      WHEN 'Okay' THEN 3
      WHEN 'Good' THEN 4
      WHEN 'Very Good' THEN 5
    END AS 'Feedback_Rating'
  FROM order_t) AS 'Rating_table'
GROUP BY quarter_number
```

Output:

Showing 4 rows

quarter_number	Avg_Rating_Per_Quarter
1	3.55
2	3.35

Output:

Showing 4 rows

quarter_number	Avg_Rating_Per_Quarter
1	3.55
2	3.35
3	2.96
4	2.4

Observations and Insights:

- The overall average ratings given by the customers is 3.135.
- From the above result, it shows that average ratings from Q1 and Q2 were above average, but decreasing from Q1 going to Q2.
- Q3 and Q4 have below overall average ratings.
- It looks like as the quarters go by, the ratings are declining, Q1 (3.55 rating) to Q4 (2.4 rating).
- From the ratings, it looks like customers are getting less and less satisfied over time.

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
SELECT quarter_number,
       ROUND(SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END) * 100.0 /
             COUNT(*), 2) AS pct_very_bad,
       ROUND(SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END) * 100.0 /
             COUNT(*), 2) AS pct_bad,
       ROUND(SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END) * 100.0 /
             COUNT(*), 2) AS pct_okay,
       ROUND(SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END) * 100.0 /
             COUNT(*), 2) AS pct_good,
       ROUND(SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) * 100.0 /
             COUNT(*), 2) AS pct_very_good
FROM order_t
GROUP BY quarter_number;
```

Output:

Result: Passed

Query: ^

```
SELECT
  quarter_number,
  ROUND(SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS pct_very_bad,
  ROUND(SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS pct_bad,
  ROUND(SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS pct_okay,
  ROUND(SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS pct_good,
  ROUND(SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS pct_very_good
FROM order_t
GROUP BY quarter_number
```

Output:

Showing 4 rows

quarter_number	pct_very_bad	pct_bad	pct_okay	pct_good	pct_very_good
1	10.97	11.29	19.03	28.71	30
2	14.89	14.12	20.23	22.14	28.63

Output:

Showing 4 rows

quarter_number	pct_very_bad	pct_bad	pct_okay	pct_good	pct_very_good
1	10.97	11.29	19.03	28.71	30
2	14.89	14.12	20.23	22.14	28.63
3	17.9	22.71	21.83	20.96	16.59
4	30.65	29.15	20.1	10.05	10.05

Observations and Insights:

- The result above shows that the customers are getting more dissatisfied over time.
- The percentage for 'Very Bad' in Q1 increases from 10.97% to 30.65% by Q4.
- The percentage for 'Very good' in Q1 decreases from 30% to 10.05% by Q4, which shows over time customers are not getting satisfied.

Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
SELECT quarter_number,  
COUNT(order_id) AS 'Total_Orders'  
FROM order_t  
GROUP BY quarter_number;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT quarter_number,  
COUNT(order_id) AS 'Total_Orders'  
FROM order_t  
GROUP BY quarter_number
```

Output:

Showing 4 rows

quarter_number	Total_Orders
1	310
2	262
3	229
4	199

Observations and Insights:

- As seen from the table above, orders steadily decline each quarter, starting from 310 orders in Q1 and only having 199 orders till Q4.
- It looks like customers are less satisfied over time.

Question 7: Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

Solution Query:

```
SELECT ROUND(SUM((vehicle_price - ((discount/100) * vehicle_price)) * quantity), 2) AS  
'Net_Revenue'  
FROM order_t;  
  
SELECT *,  
    ROUND(LAG(net_revenue, 1) OVER(ORDER BY quarter_number), 2) AS 'Prev_Qtr',  
    ROUND(100 * (net_revenue - lag(net_revenue, 1) OVER(ORDER BY quarter_number))/  
    lag(net_revenue,1) OVER(ORDER BY quarter_number), 2) AS 'Quar_O_Quar'  
FROM  
    (SELECT quarter_number,  
        ROUND(SUM((vehicle_price - ((discount/100) * vehicle_price)) * quantity), 2) AS 'Net_Revenue'  
        FROM order_t  
        GROUP BY quarter_number) AS 'New_Table';
```

Output:

Result: Passed

✓ Query 1

Query:

```
SELECT ROUND(SUM((vehicle_price - ((discount/100) * vehicle_price)) * quantity), 2) AS 'Net_Revenue'  
FROM order_t
```

Output:
Showing 1 rows

Net_Revenue
124714086.32

Result: Passed

Query 1

Query:

```
SELECT *,
       ROUND(LAG(net_revenue, 1) OVER(ORDER BY quarter_number), 2) AS 'Prev_Qtr',
       ROUND(100 * (net_revenue - lag(net_revenue, 1) OVER(ORDER BY quarter_number))/lag(net_revenue,1) OVER(ORDER BY quarter_number), 2) AS 'Quar_O_Quar'
FROM
  (SELECT quarter_number,
         ROUND(SUM((vehicle_price - ((discount/100) * vehicle_price)) * quantity), 2) AS 'Net_Revenue'
    FROM order_t
   GROUP BY quarter_number) AS 'New_Table'
```

Output:

Showing 4 rows

quarter_number	Net_Revenue	Prev_Qtr	Quar_O_Quar
1	39421580.16		
2	32715830.34	39421580.16	-17.01

Output:

Showing 4 rows

quarter_number	Net_Revenue	Prev_Qtr	Quar_O_Quar
1	39421580.16		
2	32715830.34	39421580.16	-17.01
3	29229896.19	32715830.34	-10.66
4	23346779.63	29229896.19	-20.13

Observations and Insights:

- Net Revenue generated by the company is \$124,714,086.32
- The sales are decreasing every quarter, in Q2 the sales decreased to 17%, in Q3 sales decreased by 10% and then in Q4 it decreased by 20% as compared to the previous quarters.

Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
SELECT quarter_number,  
COUNT(order_id) AS 'Total_Orders',  
ROUND(SUM((quantity * vehicle_price) * (1 - discount/100)), 2) AS 'Net_Revenue'  
FROM order_t  
GROUP BY quarter_number;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT quarter_number,  
COUNT(order_id) AS 'Total_Orders',  
ROUND(SUM((quantity * vehicle_price) * (1 - discount/100)), 2) AS 'Net_Revenue'  
FROM order_t  
GROUP BY quarter_number
```

Output:

Showing 4 rows

quarter_number	Total_Orders	Net_Revenue
1	310	39421580.16
2	262	32715830.34
3	229	29229896.19
4	199	23346779.63

Observations and Insights:

- As seen from above, there is a decline in the total number of orders per quarter from Q1 to Q4 and the net revenue also decreases over time.
- Less orders placed each quarter, is also bringing the net revenue low as well.

Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

Note: The 'discount' field in the 'order_t' table is stored as a percentage, i.e. 0.6 represents a discount of 0.6%, not 60%.

```
SELECT c.credit_card_type,  
ROUND(AVG(o.discount), 2) AS 'Avg_Discount%'  
FROM order_t AS 'o'  
    INNER JOIN customer_t AS 'c'  
    ON o.customer_id = c.customer_id  
GROUP BY c.credit_card_type  
ORDER BY ROUND(AVG(o.discount), 2) DESC;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT c.credit_card_type,  
ROUND(AVG(o.discount), 2) AS 'Avg_Discount%'  
FROM order_t AS 'o'  
    INNER JOIN customer_t AS 'c'  
    ON o.customer_id = c.customer_id  
GROUP BY c.credit_card_type  
ORDER BY ROUND(AVG(o.discount), 2) DESC
```

Output:

Showing first 10 rows out of 16 rows

credit_card_type	Avg_Discount%
laser	0.64
mastercard	0.63
visa-electron	0.62
maestro	0.62

Output:

Showing first 10 rows out of 16 rows

credit_card_type	Avg_Discount%
laser	0.64
mastercard	0.63
visa-electron	0.62
maestro	0.62
instapayment	0.62
china-unionpay	0.62
americanexpress	0.62
switch	0.61
jcb	0.61
diners-club-us-ca	0.61

Observations and Insights:

- The average discounts offered for different types of credit cards ranges between 0.61% to 0.64%, which is not very high.
- It seems like the company is not offering significant discounts depending on the credit card type, as they all are averaging around the same range.

Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
SELECT quarter_number,  
ROUND(AVG(JULIANDAY(ship_date) - JULIANDAY(order_date)), 2) AS 'Avg_Time'  
FROM order_t  
GROUP BY quarter_number;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT quarter_number,  
ROUND(AVG(JULIANDAY(ship_date) - JULIANDAY(order_date)), 2) AS 'Avg_Time'  
FROM order_t  
GROUP BY quarter_number
```

Output:

Showing 4 rows

quarter_number	Avg_Time
1	57.17
2	71.11
3	117.76
4	174.1

Observations and Insights:

- As seen from the table above, the average days between each quarter increasing significantly. In Q1 it takes about 57 days to ship the orders while in Q4 it takes about 174 days.
- Longer shipping time is be one of the factors for customers low rating as the quarters pass by, which could be the result of less orders placed, which directly correlates with low net revenue.

Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
\$124,714,086.32	1000	994	3.135
Last Quarter Revenue	Last quarter Orders	Average Days to Ship	% Good Feedback
\$23,346,779.63	199	97.96 Days	21.5%

Note: These values must be derived using SQL queries. Some of them may have already been obtained while answering previous questions.

Business Recommendations

- There are more orders (1000) than customers (994), which means there were multiple orders placed by the same customers.
- As you saw from above, number of orders decreased every quarter. Q1 has 310 orders placed and Q4 had 199 orders placed.
- % Good Feedback is low, which means customers are not satisfied.
- The recommendations for the business is to decrease average shipping time drastically to about 60 days. This is a biggest factor in customers dissatisfaction.
- There has to be attention given to logistics and supply chain management immediately and increase the average days to ship the car to the customers, which will impact the revenue and ratings of the business.